Exercises

Level - 1

(Problems Based on Fundamentals)

ABC of Tangents and Normals

1. Find the slope of the tangent to the curve $y = x^3 + 3x^2 + 3x - 10$ at $x = 2$.

2. Find the slope of the normal to the curve $y = x^3 + 1$ at $x = 2$.

3. If the slope of the curve $y = \frac{ax}{b - x}$ at the point (1, 1) be 2. Find the value of $a + b + 10$.

4. Find the equation of the tangent to the curve $y = e^{2x}$ at $x = 0$

5. Find the equation of the tangent to the curve $y = \sqrt[3]{x - 1}$ at $x = 1$.

6. Find the equation of the tangent to the curve $y = be^{-x/a}$ at the point where it crosses the y-axis.

7. Find the equation of the tangent to the curve $y = \frac{4}{x^2 + 2}$ at $x = 0$.

8. Find the equation of the normal to the curve $x^3 + y^3 = txy$ at (3, 3).

9. Find the equation of the normal to the curve $y = 3x^2 + 2 \sin x + 4 \text{ ccos } x + 10$ at $x = 0$.

10. Find the equation of the normal to the curve $x + y = x^y$, where it curts the x-axis.

11. Find the equation of the normal to the curve $y = |x^2 - |x||$ at $x = -2$.

12. Find the equation of the tangents drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$ from point (1, 2).

13. Find the equation of the normal to the curve $x^2 = 4y$, which passes through the point (1, 2)

Vertical Tangent

14. Show that the curve

$y = x - e^{xy}$ has a vertical tangent at (1, 0).

15. The curve $x + y - \log(x + y) = 2x + 5$ has a vertical tangent at the point (p, q), then find the value of $p + q + 10$.

Horizontal Tangent

16. Find the points on the curve

$y = 2x^3 + 3x^2 - 12x + 1$ where the tangent is horizontal.

17. Find the points on the curve $y = x^3 - x^2 - x + 3$ where the tangents are parallel to x-axis.

Equally Inclined with the Axes

18. At what points on the curve $y = \frac{x^3}{3} + \frac{x^2}{2}$, the tangents make equal angles with the co-ordinate axes?

Equation of Tangent and Normal to a 2nd Degree Curve

19. Find the equation of the tangent to the curve $y = 4ax$ at 't'.
20. Find the equation of the tangent to the curve $x^2 + y^2 + x + y = 0$ at $(1, -1)$
21. Find the equation of the normal to the curve $x^2 + y^2 = 10$ at $(3, 1)$.
22. Find the equation of the normal to the curve $x^2 + y^2 + 4x + 6y + 9 = 0$ at $(-4, -3)$.
23. Find the equation of the normal to the curve $\frac{x^2}{9} + \frac{y^2}{4} = 2$ at $(3, 2)$.
24. Find the number of tangents to the curve $y^2 - 2x^2 - 4y + 8 = 0$, which pass through the point $(1, 2)$.
25. Any tangent at a point $P(x, y)$ to the ellipse $\frac{x^2}{8} + \frac{y^2}{18} = 1$ meets the co-ordinate axes in the points A and B such that the area of the triangle OAB is least, then find the point P.

Length of Intercepts of the Tangents by the Axes

26. If the tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at any point on it cuts the axes OX, OY respectively, then prove that OP + OQ = a.
27. Show that the x-intercept of the tangent at an arbitrary point of the curve $\frac{a}{x^2} + \frac{b}{y^2} = 1$ is proportional to the cube of the abscissa of the point of tangency.

Tangent to the Curve at the Origin

28. Find the tangent to the curve $x^3 + 3xy + y^3 + x^2y = 0$ at the origin.
29. Find the tangent to the curve $ax^2 + 2hxy + by^2 + ax + by = 0$ at the origin.
30. Find the tangent to the curve $(x^4 + y^4)^2 = 2013(x^2 - y^2)$ at the origin.
31. Find the tangent to the curve $x^5 + y^5 + 2010x^2 - 2011y^2 + 2012x - 2013y = 0$ at the origin.

Angle between two Curves

32. Find the angle of intersection of the curves $x^2 = y$ and $y^2 = x$.
33. Find the angle of intersection of the curves $y = 4 - x^2$ and $y = x^2$.
34. Find the angle between the curves $y^2 = 4x$ and $y = e^{-x/2}$.
35. Find the acute angle between the curves $y = \sin x$ and $y = \cos x$.
36 Find the angle between the curves $2y^2 = x^3$ and $y^2 = 32x$.
37. Prove that the curves $y^2 = 4x$ and $x^2 + y^2 - 6y + 1 = 0$ touch each other at the point $(1, 2)$.
38. Prove that the curves $y = 6 - x + x^2$ and $y = (x - 1)(x + 2)$ touch each other at $(2, 4)$.
39. Prove that the curves $x = y$ and $xy = k$ cut at right angles if $8k^2 = 1$.
40. Find the values of a if the curves $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ and $y^3 = 16x$ cut each other orthogonally.
41. Find the acute angle between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their points of intersection.
42. Find the angle of intersection of the curves $y = [|\sin x| + |\cos x|]$, where [,] = G.I.F and $x^2 + y^2 = 5$.

Shortest Distance between two Curves

43. Find the shortest distance between the line $y = x - 2$ and the curve $y = x^2 + 3x + 2$
44. Find the shortest distance between the curves $y^2 = 4x$ and $x^2 + y^2 - 12x + 31 = 0$.
45. Find the shortest distance between the curves $y^2 = x^3$ and $9x^2 + 9y^2 - 30y + 16 = 0$.
46. Find a point on the curve $x^2 + 2y^2 = 6$ whose distance from the line $x + y = 7$ is minimum.
47. Find the least distance between any two points of the curves $y = 3^x$ and $\log_3 x$.

Common Tangent between two Curves

48. Find the common tangent to the curves $y = x^2 + x + 1$ and $y = x^2 - 5x + 6$.
49. Find the equation of the common tangent to the curves $y = 3x^2$ and $y = 2x^3 + 1$.
50. Find the equation of the common tangent to the curves $y = 6 - x - x^2$ and $xy = x + 3$.

Tangent at a Point Intersect the Curve Again 2nd Point

51. If the tangent at $P(1, 1)$ to the curve $y^2 = x(2 - x)^2$ meets the curve again at Q, then find the co-ordinates of Q.
52. If the tangent at P to the curve $y^2 = x^3$ intersects the curve again at Q and the straight line OP, OQ makes

angles α, β with the x-axis, where O igin, then find the value of $\left(\frac{\tan\alpha}{\tan\beta} + 2013\right)$.

53. If the tangent at a variable point P on the rve $y = x^2 - x^3$ meets it again at Q. Show that the locus the middle point of PQ is $y = 1 - 9x + 28x^2 - 28x^3$.
54. A curve is given by the equations $x = \sec^2\theta$ and $y = \cot\theta$. If the tangent at P, where $\theta = \frac{\pi}{4}$ meets the curve again at Q. Find the length of PQ.

Lengths of Tangent, Normal, Sub-tangent and Sub-normal

55. Find the lengths of the tangent, sub-tangent, normal and sub-normal to the curve $y^2 = 4ax$ at the point $P(at^2, 2at)$.
56. Prove that the length of the sub-tangent at any point to the curve $y = be^{x/a}$ is always constant.
57. Find the length of the tangent and normal to the curves $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ at the point, where $\theta = \frac{\pi}{2}$.
58. Find the length of the sub-normal to the curve $y^2 = x^3$ at $(4, 8)$.
59. Find the length of the sub-tangent to the curve $y = \frac{c}{2}\left(e^{\frac{x}{c}} + e^{-\frac{x}{c}}\right)$ at any point (x, y).

Solutions:

3 Since the point (1, 1) lies on the curve,

so $1 = \frac{a}{b-1}$

$\Rightarrow \quad a = b - 1$

Also, $\frac{dy}{dx} = \frac{ab}{(b-x)^2}$

Now, $\left(\frac{dy}{dx}\right)_{(1,1)} = 2$

$\Rightarrow \quad \frac{ab}{(b-1)^2} = 2$

$\Rightarrow \quad a(a+1) = 2a^2$

$\Rightarrow \quad a^2 + a = 2a^2$

$\Rightarrow \quad a^2 = a$

$\Rightarrow \quad a = 0, 1$

when $a = 0, b = 1$

Then $a + b + 10 = 0 + 1 + 10 = 11$

when $a = 1, b = 2$

Then $a + b + 10 = 1 + 2 + 10 = 13$.

4. when $x = 0, y = 1$

Thus, the point is (0, 1)

Now, $\frac{dy}{dx} = 2.e^{2x}$

Thus, $m = \left(\frac{dy}{dx}\right)_{x=0} = 2.1 = 2$

Hence, the equation of the tangent is

$y - 1 = 2(x - 0)$

$\Rightarrow \quad y = 2x + 1$

5. when $x = 1, y = 0$

Thus, the point is (1, 0)

Now, $\frac{dy}{dx} = \frac{1}{3}(x-1)^{-\frac{2}{3}}$

Thus, $m = \left(\frac{dy}{dx}\right)_{x=1} = \infty$

Hence, the equation of the tangent to the curve is

$y - 0 = \infty(x - 1)$

$\Rightarrow \quad (x-1) = \frac{y}{\infty} = 0$

$\Rightarrow \quad x = 1$

6. The equation of the given curve is $y = be^{-x/a}$

put $x = 0$, then $y = b$

So, the point is $(0, b)$

Now, $\frac{dy}{dx} = -\frac{b}{a}e^{-x/a}$

Thus, $m = \left(\frac{dy}{dx}\right)_{(0,b)} = -\frac{b}{a}$

Hence, the equation of the tangent is

$y - b = -\frac{b}{a}(x - 0)$

$\Rightarrow \quad ay - ab = -bx$

$\Rightarrow \quad bx + ay = ab$

$\Rightarrow \quad \frac{bx}{ab} + \frac{ay}{ab} = \frac{ab}{ab}$

$\Rightarrow \quad \frac{x}{a} + \frac{y}{b} = 1$

7. when $x = 0, y = 2$

So, the point is (0, 2)

Now, $\frac{dy}{dx} = -\frac{8x}{(x^2+2)^2}$

Thus, $m = \left(\frac{dy}{dx}\right)_{x=0} = -\frac{8}{4} = -2$

Hence, the equation of the tangent is

$y - 2 = -2(x - 0)$

$\Rightarrow \quad y = 2 - 2x$.

8. The given curve is $x^3 + y^3 = 3xy$

$\Rightarrow \quad 3x^2 + 3y^2\frac{dy}{dx} = 6\left(x\frac{dy}{dx} + y \cdot 1\right)$

$\Rightarrow \quad (y^2 - 2x)\frac{dy}{dx} = (2y - x^2)$

$\Rightarrow \quad \frac{dy}{dx} = \left(\frac{2y - x^2}{y^2 - 2x}\right)$

Thus, $m = \left(\frac{dy}{dx}\right)_{(3,3)} = \frac{6-9}{9-6} = -1$

Hence, the equation of the normal is

$y - 3 = 1(x - 3)$

$\Rightarrow \quad x - y = 0$

9. when $x = 0, y = 14$

Hence, the point is (0, 14)

Now, $\frac{dy}{dx} = 6x + 2\cos x - 4\sin x$

Thus, $m = \left(\frac{dy}{dx}\right)_{x=0} = 2$

Hence, the equation of the normal is

$y - 14 = -\frac{1}{2}(x - 0)$

$\Rightarrow \quad 2y - 28 = -x$

$\Rightarrow \quad x + 2y = 28$.

10. The equation of the given curve is $x + y = x^y$ put $y = 0$, then $x = 1$.
So, the point is (1, 0)
Now, $x + y = x^y$
$\Rightarrow \log(x + y) = y\log x$
$\Rightarrow \frac{1}{(x+y)}\left(1 + \frac{dy}{dx}\right) = y\cdot\frac{1}{x} + \log x\cdot\frac{dy}{dy}$
put $x = 1, y = 0, \left(1 + \frac{dy}{dx}\right) = 0$
$\Rightarrow \left(\frac{dy}{dx}\right) = -1$
$\Rightarrow$ Slope of normal = 1
Hence, the equation of the normal is
$y - 0 = 1\ (x - 1)$
$\Rightarrow y = x - 1.$

11. The equation of the given curve is
$y = |x^2 - |x||$
$\Rightarrow y = |x^2 - (-x)|$ (since at $x = -2$, $|x| = -x$)
$\Rightarrow y = |x^2 + x|$
$\Rightarrow y = x^2 + x$ (at $x = -2$, $x^2 + x > 0$)
when $x = -2, y = 2$
So, the point is $(-2, 2)$
Now, $\frac{dy}{dx} = 2x + 1$
Thus, $m = \left(\frac{dy}{dx}\right)_{x=-2} = -3$
So, slope of the normal = 1/3
Hence, the equation of the normal is
$y - 2 = \frac{1}{3}(x + 2)$
$\Rightarrow 3y - 6 = x + 2$
$\Rightarrow 3y = x + 8$

12. The equation of the given curve is
$y^2 - 2x^3 - 4y + 8 = 0$
$\Rightarrow \frac{dy}{dx} = \frac{3x^2}{y - 2}$
Let the point (α, β) lies on the curve
Thus, $m = \left(\frac{dy}{dx}\right)_{(\alpha,\beta)} = \frac{3\alpha^2}{\beta - 2}$
Therefore, the equation of the tangent at
(α, β) is $y - \beta = \left(\frac{3\alpha^2}{\beta - 2}\right)(x - \alpha)$ ...(1)
which is passing through (1, 2)
So, $(2 - \beta) = \left(\frac{3\alpha^2}{\beta - 2}\right)(1 - \alpha)$ ...(2)
Also, the point (α, β) lies on the curve
$y^2 - 2x^3 - 4y + 8 = 0$
So, $\beta^2 - 2\alpha^2 - 4\beta + 8 = 0$ ...(3)
From (2) and (3), we get,
$2(\alpha^3 - 2) = 3\alpha^2(\alpha - 1)$
$\Rightarrow \alpha^3 - 3\alpha^2 + 4 = 0$
$\Rightarrow \alpha = 2$
when $\alpha = 2, \beta = \pm 2\sqrt{3}$
Hence, the equation of the tangents are
$(y - (2 \pm 2\sqrt{3})) = \pm 2\sqrt{3}(x - 2)$

13. The equation of the given curve is $x^2 = 4y$
$\Rightarrow 2x = 4\frac{dy}{dx}$
$\Rightarrow \frac{dy}{dx} = \frac{x}{2}$
Let the point on the given curve be (α, β)
Now, $m = \left(\frac{dy}{dx}\right)_{(\alpha,\beta)} = \frac{\alpha}{2}$
Therefore the equation of normal at (α, β) is
$y - \beta = -\frac{2}{\alpha}(x - \alpha)$ ...(1)
which is passing through (1, 2).
So, $2 - \beta = -\frac{2}{\alpha}(1 - \alpha)$ ...(2)
Also, the point (α, β) lies on the curve
$\alpha^2 = 4\beta$ ...(3)
Solving (2) and (3), we get, $\alpha = 2, \beta = 1$
Hence, the equation of the normal is
$y - 1 = -\frac{2}{2}(x - 2)$
$\Rightarrow y - 1 = -x + 2$
$\Rightarrow x + y = 3.$

14. The equation of the given curve is $y = x - e^{xy}$
$\Rightarrow \frac{dy}{dx} = 1 - e^{xy}\left(y\cdot 1 + x\cdot\frac{dy}{dx}\right)$
$\Rightarrow (1 - xe^{xy})\frac{dy}{dx} = (ye^{xy} - 1)$
$\Rightarrow \frac{dy}{dx} = \frac{ye^{xy} - 1}{1 - xe^{xy}}$
For a vertical tangent, $\frac{dx}{dy} = 0$
$\Rightarrow 1 - xe^{xy} = 0$
$\Rightarrow x = 1, y = 0$
Therefore, the curve $y = x - e^{xy}$ has a vertical tangent at (1, 0).

15. The given curve is $x + y - \log(x + y) = 2x + 5$
$\Rightarrow 1 + \frac{dy}{dx} - \frac{1}{x+y}\left(1 + \frac{dy}{dx}\right) = 2$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{x+y+1}{x+y-1}$$

Now, $\left(\frac{dy}{dx}\right)_{(p,q)} = \frac{p+q+1}{p+q-1}$

Since, the tangent is vertical, so $\frac{dx}{dy} = 0$

$\Rightarrow \quad p + q = 1$

The value of $p + q + 10 = 11$.

16. Since the curve has horizontal tangent, so $\frac{dx}{dy} = 0$

$\Rightarrow \quad 6x^2 + 6x - 12 = 0$

$\Rightarrow \quad x^2 + x - 2 = 0$

$\Rightarrow \quad (x+2)(x-1) = 0$

$\Rightarrow \quad x = 1, -2$

When $\quad x = 1, y = 2 + 3 - 12 + 1 = -6$

So, the point is $(1, -6)$

when $x = -2$, $y = 16 + 12 - 26 + 1 = 3$

So, the point is $(-2, 3)$

Hence, the points are $(1, -6)$ and $(-2, 3)$.

17. Since the tangents are parallel to x-axis, so $\frac{dy}{dx} = 0$

$\Rightarrow \quad 3x^2 - 2x - 1 = 0$

$\Rightarrow \quad (3x+1)(x-1) = 0$

$\Rightarrow \quad x = 1, -1/3$

when $\quad x = 1, y = 1 - 1 - 1 + 3 = 2$

when $\quad x = -1/3, y = 70/27$

Hence, the points are $(1, 2)$ and $(-1/3, 70/27)$

18. The given curve is $y = \frac{x^3}{3} + \frac{x^2}{2}$

$\Rightarrow \quad \frac{dy}{dx} = 2x^2 + x$

Since the tangents make equal angles with the axes, so $\frac{dy}{dx} = \pm 1$

$\Rightarrow \quad 2x^2 + x = \pm 1$

$\Rightarrow \quad 2x^2 + x - 1 = 0, 2x^2 + x + 1 = 0$

$\Rightarrow \quad (2x-1)(x+1) = 0$

$\Rightarrow \quad x = 1/2, -1$

when $\quad x = 1/2, y = 5/24$

and when $x = -1$, $y = 1/6$

Hence, the points are $(1/2, 5/24)$ & $(-1, 1/6)$

19. The equation of the tangent to the curve

$y^2 = 4ax$ is $yy_1 = 2a(x + x_1)$

$\Rightarrow \quad y \cdot 2at = 2a(x + at^2)$

$\Rightarrow \quad yt = x + at^2$

20. The equation of the tangent to the curve $x^2 + y^2 + x + y = 0$ is

$$xx_1 + yy_1 + \left(\frac{x+x_1}{2}\right) + \left(\frac{y+y_1}{2}\right) = 0$$

$$\Rightarrow \quad x \cdot 1 - y \cdot 1 + \left(\frac{x+1}{2}\right) + \left(\frac{y-1}{2}\right) = 0$$

$\Rightarrow \quad 2x - 2y + x + 1 + y - 1 = 0$

$\Rightarrow \quad 3x - y = 0$

21. Equation of the normal to the curve

$x^2 + y^2 = 10$ is $\frac{x}{x_1} = \frac{y}{y_1}$

$\Rightarrow \quad \frac{x}{3} = \frac{y}{1}$

$\Rightarrow \quad x - 3y = 0$

22. The equation of the normal to the curve $x^2 + y^2 + 4x + 6y + 9 = 0$ at (x_1, y_1) is

$$\frac{x - x_1}{x_1 + 2} = \frac{y - y_1}{y_1 + 3}$$

$$\Rightarrow \quad \frac{x+4}{-4+2} = \frac{y+3}{-3+3}$$

$\Rightarrow \quad y + 3 = 0.$

23. Equation of the tangent to the curve is

$$\frac{xx_1}{9} + \frac{yy_1}{4} = 2$$

$$\Rightarrow \quad \frac{3x}{9} + \frac{2y}{4} = 2$$

$$\Rightarrow \quad \frac{x}{3} + \frac{y}{2} = 2$$

$\Rightarrow \quad 2x + 3y = 6$

Thus, slope of the tangent is $= -2/3$

Therefore, the slope of the normal is $= 3/2$

Equation of the normal to the curve at $(3, 2)$ is

$(y - 2) = \frac{3}{2}(x - 3)$

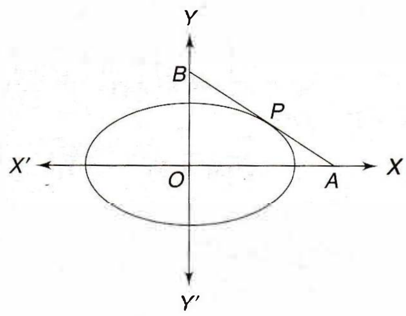
$\Rightarrow \quad 2y - 4 = 3x - 9$

$\Rightarrow \quad 3x - 2y = 5.$

24. Clearly, the point $(1, 2)$ lies outside of the curve $y^2 - 2x^2 - 4y + 8 = 0$. (as $4 - 2 - 4 + 8 = 12 - 6 = 6 > 0$)

Since the point lies outside of the given curve, so the number of tangent will be 2.

25.



Let the point P be $(2\sqrt{2}\cos\theta, 3\sqrt{2}\sin\theta)$

Equation of the tangent at P is

$$\frac{xx_1}{8} + \frac{yy_1}{18} = 1$$

$$\Rightarrow \quad \frac{x \cdot 2\sqrt{2}\cos\theta}{8} + \frac{y \cdot 3\sqrt{2}\sin\theta}{18} = 1$$

$$\Rightarrow \quad \frac{x}{2\sqrt{2}\sec\theta} + \frac{y}{3\sqrt{2}\operatorname{cosec}\theta} = 1$$

Thus, the co-ordinates of A and B are

$$(2\sqrt{2}\sec\theta, 0) \text{ and } (0, 3\sqrt{2}\operatorname{cosec}\theta)$$

Now, $\quad ar(\Delta OAB) = \frac{1}{2} \times OA \times OB$

$$= \frac{1}{2} \times 2\sqrt{2}\sec\theta \times 3\sqrt{2}\operatorname{cosec}\theta$$

$$= \frac{12}{2\sin\theta\cos\theta}$$

$$= \frac{12}{\sin 2\theta}$$

The area of the triangle is maximum, when $\sin 2\theta = 1$

$$\Rightarrow \quad 2\theta = \frac{\pi}{2}$$

$$\Rightarrow \quad \theta = \frac{\pi}{4}$$

Hence, the point P is (2, 3).

26. Let any point on the curve be $M(x_1, y_1)$

The equation of the given curve is

$$\sqrt{x} + \sqrt{y} = \sqrt{a}$$

$$\Rightarrow \quad \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_M = -\frac{\sqrt{y_1}}{\sqrt{x_1}}$$

Now, $\quad x\text{-intercept} = OP = x_1 - \frac{y_1}{dy/dx}$

$$= x_1 - \frac{y_1}{\left(-\frac{\sqrt{y_1}}{\sqrt{x_1}}\right)} = x_1 + \sqrt{x_1 y_1}$$

$$y\text{-intercept} = y_1 - x_1\frac{dy}{dx}$$

$$= y_1 - x_1\left(-\frac{\sqrt{y_1}}{\sqrt{x_1}}\right) = y_1 + \sqrt{x_1 y_1}$$

Thus, $OP + OQ$

$$= x_1 + \sqrt{x_1 y_1} + y_1 + \sqrt{x_1 y_1}$$

$$= x_1 + y_1 + 2\sqrt{x_1 y_1}$$

$$= (\sqrt{x_1} + \sqrt{y_1})^2$$

$$= (\sqrt{a})^2 = a$$

27. Let the point on the curve be $P(\alpha, \beta)$

The equation of the given curve is $\frac{a}{x^2} + \frac{b}{y^2} = 1$

$$\Rightarrow \quad -\frac{2a}{x^3} - \frac{2b}{y^3}\frac{dy}{dx} = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = -\frac{ay^3}{bx^3}$$

Now, $\quad \left(\frac{dy}{dx}\right)_P = -\frac{a\beta^3}{b\alpha^3}$

Therefore, x-intercept

$$= x - \frac{y}{dy/dx}$$

$$= \alpha - \frac{\beta}{\left(-\frac{a\beta^3}{b\alpha^3}\right)} = \alpha + \frac{b\alpha^3}{a\beta^2}$$

$$= \alpha + \frac{\alpha^3}{a} \times \frac{b}{\beta^2}$$

$$= \alpha + \frac{\alpha^3}{a} \times \left(1 - \frac{a}{\alpha^2}\right)$$

$$= \alpha + \frac{\alpha^3}{a} - \alpha$$

$$= \frac{\alpha^3}{a}$$

$\Rightarrow \quad$ x-intercept is proportional to α^3.

28. The equation of the tangent to the origin is

$$xy = 0.$$

$$\Rightarrow \quad x = 0 \text{ and } y = 0$$

29. The equation of the tangent at the origin is

$$ax + by = 0.$$

30. The equation of the tangent to the curve at the origin is

$$x^2 - y^2 = 0$$

$$\Rightarrow \quad x + y = 0 \text{ and } x - y = 0.$$

31. The equation of the tangent to the curve at the origin is $2012x - 2013y = 0$.

32. The given curves are $x^2 = y$ and $y^2 = x$

We have, $x = x^4$

$$\Rightarrow \quad x(1 - x^3) = 0$$

$$\Rightarrow \quad x = 0, 1$$

when $x = 0$, $y = 0$ and when $x = 1$, $y = 1$

So, the point of intersections are (0, 0), (1, 1)

At the point of intersection (0, 0)

Now, $\quad y = x^2$

$$\Rightarrow \quad \frac{dy}{dx} = 1$$

$$m_1 = \left(\frac{dy}{dx}\right)_{(0,0)} = 1$$

Also, $y^2 = x$

$\Rightarrow \quad 2y\dfrac{dy}{dx} = 1$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{1}{2y}$

$$m_2 = \left(\frac{dy}{dx}\right)_{(0,0)} = \frac{1}{0} = \infty$$

Let θ be the angle between them

Then, $\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right| = \infty$

$\Rightarrow \quad \theta = \dfrac{\pi}{2}$

Now, consider the point of intersection (1, 1)

$$m_1 = \left(\frac{dy}{dx}\right)_{(1,1)} \text{ and } m_2 = \left(\frac{dy}{dx}\right)_{(1,1)} = \frac{1}{2}$$

Let ϕ be the angle between them

Then $\tan(\varphi) = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\dfrac{1 - \frac{1}{2}}{1 + 1\cdot\frac{1}{2}}\right| = \dfrac{1}{3}$

$\Rightarrow \quad \varphi = \tan^{-1}\left(\dfrac{1}{3}\right).$

33. The given curves are $y = 4 - x^2$ and $y = x^2$
On solving, we get, $x^2 = 4 - x^2$

$\Rightarrow \quad x = \pm\sqrt{2}$

when $x = \pm\sqrt{2}$, $y = 2$

So, the points of intersections are

$(\sqrt{2}, 2)$ & $(-\sqrt{2}, 2)$

Now, $y = 4 - x^2$

$\Rightarrow \quad \dfrac{dy}{dx} = -2x$

$$m_1 = \left(\frac{dy}{dx}\right)_{(\sqrt{2},\,2)} = -2\sqrt{2}$$

Also, $y = x^2$

$\Rightarrow \quad \dfrac{dy}{dx} = 2x$

$$m_2 = \left(\frac{dy}{dx}\right)_{(\sqrt{2},\,2)} = 2\sqrt{2}$$

Let θ be the angle between them

Then,

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{2\sqrt{2} + 2\sqrt{2}}{1 - 8}\right| = \left(\frac{4\sqrt{2}}{7}\right)$$

$\Rightarrow \quad \theta = \tan^{-1}\left(\dfrac{4\sqrt{2}}{7}\right)$

Similarly, we can find, $\varphi = \tan^{-1}\left(\dfrac{4\sqrt{2}}{7}\right)$

34. The given curves are $y^2 = 4x$ and $y = e^{-x/2}$
Let the point of intersection be (x_1, y_1)

Now, $y^2 = 4x$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{2}{y}$

$\Rightarrow \quad m_1 = \left(\dfrac{dy}{dx}\right)_{(x_1,\,y_1)} = \dfrac{2}{y_1}$

Also, $y = e^{-x/2}$

$$\frac{dy}{dx} = e^{-x/2} \times -\frac{1}{2}$$

$\Rightarrow \quad m_2 = \left(\dfrac{dy}{dx}\right)_{(x_1,\,y_1)} = -\dfrac{1}{2} \times e^{\frac{-x_1}{2}} = -\dfrac{1}{2}y_1$

Let θ be the angle between them

Then $\tan\theta = \left|\dfrac{m_2 - m_1}{1 + m_1 m_2}\right|$

$\Rightarrow \quad \tan\theta = \left|\dfrac{-\frac{y_1}{2} - \frac{2}{y_1}}{1 + \left(\frac{-y_1}{2} \times \frac{2}{y_1}\right)}\right| = \infty$

$\Rightarrow \quad \theta = \dfrac{\pi}{2}$

35. The given curves are $y = \sin x$ and $y = \cos x$.
On solving, we get, the point of intersection is

$\left(\dfrac{\pi}{4}, \dfrac{1}{\sqrt{2}}\right).$

Now, $y = \sin x$

$\Rightarrow \quad \dfrac{dy}{dx} = \cos x$

$\Rightarrow \quad m_1 = \left(\dfrac{dy}{dx}\right)_{\left(\frac{\pi}{4},\,\frac{1}{\sqrt{2}}\right)} = \dfrac{1}{\sqrt{2}}$

Also, $y = \cos x$

$\Rightarrow \quad \dfrac{dy}{dx} = -\sin x$

$\Rightarrow \quad m_2 = \left(\dfrac{dy}{dx}\right)_{\left(\frac{\pi}{4},\,\frac{1}{\sqrt{2}}\right)} = -\dfrac{1}{\sqrt{2}}$

Let θ be the angle between them

Then $\tan\theta = \left|\dfrac{m_2 - m_1}{1 + m_1, m_2}\right|$

$\Rightarrow \quad \tan\theta = \left|\dfrac{-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}}{1 - \frac{1}{2}}\right| = 2\sqrt{2}$

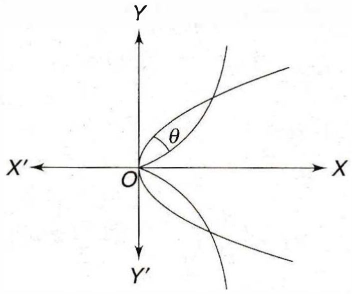
$\Rightarrow \quad \theta = \tan^{-1}(2\sqrt{2})$

36. The given curves are $2y^2 = x^3$ and $y^2 = 32x$

On solving we get, the point of intersections are $O(0, 0)$, $P(8, 16)$ and $Q(8, -16)$.

From the diagram, it is clear that, the angle of intersection at (0, 0) is $\frac{\pi}{2}$.

Now, $\quad 2y^2 = x^3$



From the diagram, it is clear that, the angle of intersection at (0, 0) is $\frac{\pi}{2}$.

Now, $\quad 2y^2 = x^3$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{3x^2}{4y}$$

$$\Rightarrow \quad m_1 = \left(\frac{dy}{dx}\right)_P = \frac{3 \times 64}{4 \times 16} = 3$$

Also, $\quad y^2 = 32x$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{32}{2y} = \frac{16}{y}$$

$$\Rightarrow \quad m_2 = \left(\frac{dy}{dx}\right)_P = \frac{16}{16} = 1$$

Let θ be the angle between them

Then, $\quad \tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$

$$\Rightarrow \quad \tan\theta = \left|\frac{1-3}{1+1.3}\right| = \frac{1}{2}$$

$$\Rightarrow \quad \theta = \tan^{-1}\left(\frac{1}{2}\right)$$

Thus, the angle of intersection at P and Q is $\tan^{-1}\left(\frac{1}{2}\right)$.

37. The given curves are $y^2 = 4x$ and

$x^2 + y^2 - 6x + 1 = 0$

Now, $\quad y^2 = 4x$

$$\Rightarrow \quad 2y\frac{dy}{dx} = 4$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{2}{y}$$

$$\Rightarrow \quad m_1 = \left(\frac{dy}{dx}\right)_{(1,2)} = \frac{2}{2} = 1$$

Also, $\quad x^2 + y^2 - 6x + 1 = 0$

$$\Rightarrow \quad 2x + 2y \cdot \frac{dy}{dx} - 6 = 0$$

$$x + y\frac{dy}{dx} - 3 = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{3-x}{y}$$

$$\Rightarrow \quad m_2 = \left(\frac{dy}{dx}\right)_{(1,2)} = \frac{3-1}{2} = 1$$

Let θ be the angle between them

Then $\quad \tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 \cdot m_2}\right| = \left|\frac{1-1}{1+1.1}\right| = 0$

$$\Rightarrow \quad \theta = 0$$

Hence, the curves touch each other.

38. The given curves are $y = 6 - x + x^2$

and $\quad y(x-1) = x + 2$.

Now, $\quad y = 6 - x + x^2$

$$\frac{dy}{dx} = -1 + 2x$$

$$m_1 = \left(\frac{dy}{dx}\right)_{(2,4)} = -1 + 4 = 3$$

Also, $\quad y = (x-1)(x+2)$

$$\Rightarrow \quad \frac{dy}{dx} = x + 2 + x - 1 = 2x + 1$$

$$\Rightarrow \quad m_2 = \left(\frac{dy}{dx}\right)_{(2,4)} = 3$$

Let θ be the angle between them

Then, $\quad \tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 \cdot m_2}\right|$

$$\Rightarrow \quad \tan = \theta = \left|\frac{3-3}{1+9}\right| = 0$$

$$\Rightarrow \quad \theta = 0$$

Hence, the curves touch each other

39. The given curves are $x = y^2$ and $x\,y = k$

On solving we get, $y = k^{1/3}$, $x = k^{2/3}$

So, the point of intersection is $P\,(k^{2/3}, k^{1/3})$

Now, $\quad y^2 = x$

$$\Rightarrow \quad 2y\frac{dy}{dx} = 1$$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{1}{2y}$

$\Rightarrow \quad m_1 = \left(\dfrac{dy}{dx}\right)_P = \dfrac{1}{2k^{1/3}}$

Also, $\quad xy = k \Rightarrow \dfrac{dy}{dx} = -\dfrac{k}{x^2}$

$\Rightarrow \quad m_2 = \left(\dfrac{dy}{dx}\right)_P = \dfrac{k}{k^{4/3}} = -\dfrac{1}{k^{1/3}}$

since the given curves cut at right angles, so

$m_1 \times m_2 = -1$

$\Rightarrow \quad \dfrac{1}{2k^{1/3}} \times -\dfrac{1}{k^{1/3}} = -1$

$\Rightarrow \quad \dfrac{1}{2k^{2/3}} = 1$

$\Rightarrow \quad \dfrac{1}{8k^2} = 1$

$\Rightarrow \quad 8k^2 = 1$

40. The given curves are $\dfrac{x^2}{a^2} + \dfrac{y^2}{4} = 1$ and $y^3 = 16x$

Let the point of intersection be (x_1, y_1)

Now, $\quad \dfrac{x^2}{a^2} + \dfrac{y^2}{4} = 1$

$\Rightarrow \quad \dfrac{2x}{a^2} + \dfrac{2y}{4}\dfrac{dy}{dx} = 0$

$\Rightarrow \quad \dfrac{x}{a^2} + \dfrac{y}{2} \cdot \dfrac{dy}{dx} = 0$

$\Rightarrow \quad \dfrac{dy}{dx} = -\dfrac{2x}{a^2 y}$

$\Rightarrow \quad m_1 = \left(\dfrac{dy}{dx}\right)_P = -\dfrac{2x_1}{a^2 y_1}$

Also, $\quad y^3 = 16x$

$\Rightarrow \quad 3y^2 \dfrac{dy}{dx} = 16$

$\Rightarrow \quad \dfrac{dy}{dx} = \dfrac{16}{3y^2}$

$\Rightarrow \quad m_2 = \left(\dfrac{dy}{dx}\right)_P = \dfrac{16}{3{y_1}^2}$

Since two curves are orthogonal, so

$m_1 \times m_2 = -1$

$\Rightarrow \quad -\dfrac{2x_1}{a^2 y_1} \times \dfrac{16}{3{y_1}^2} = -1$

$\Rightarrow \quad \dfrac{32x_1}{3a^2{y_1}^3} = 1$

$\Rightarrow \quad \dfrac{32x_1}{3a^2 \cdot 16x_1} = 1$

$\Rightarrow \quad a^2 = \dfrac{2}{3}$

$\Rightarrow \quad a = \pm\sqrt{\dfrac{2}{3}}$

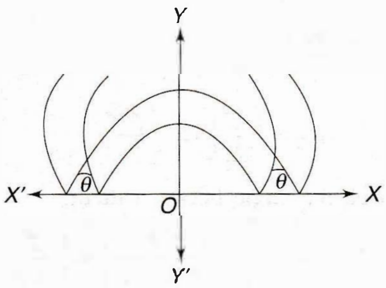
41. The given curves are $y = |x^2 - 1|$ and

$y = |x^2 - 3|$

On solving, we get, $x^2 - 1 = -x^2 + 3$

$\Rightarrow \quad 2x^2 = 4$

$\Rightarrow \quad x^2 = 2$



$\Rightarrow \quad x = \pm\sqrt{2}$

when $\quad x = \pm\sqrt{2}$, then $y = 2 - 1 = 1$

so, the point of intersection is $(\pm\sqrt{2}, 1)$

Now, consider the point of intersection is $P(\sqrt{2}, 1)$

Now, $\quad y = |x^2 - 1|$

$\Rightarrow \quad y = x^2 - 1$

$\Rightarrow \quad \dfrac{dy}{dx} = 2x$

$\Rightarrow \quad m_1 = \left(\dfrac{dy}{dx}\right)_P = 2\sqrt{2}$

Also, $\quad y = |x^2 - 3|$

$\Rightarrow \quad y = -x^2 + 3$

$\Rightarrow \quad \dfrac{dy}{dx} = -2x$

$\Rightarrow \quad m_2 = \left(\dfrac{dy}{dx}\right)_P = -2\sqrt{2}$

Let θ be the angle between them

Then $\quad \tan\theta = \left|\dfrac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\dfrac{-2\sqrt{2} - 2\sqrt{2}}{1 + 2\sqrt{2} \cdot (-2\sqrt{2})}\right|$

$\Rightarrow \quad \tan\theta = \dfrac{4\sqrt{2}}{7}$

$\Rightarrow \quad \theta = \tan^{-1}\left(\dfrac{4\sqrt{2}}{7}\right).$

42. We have, $y = [|\sin x| + |\cos x|]$

$\Rightarrow \quad y = 1$, since $1 \le |\sin x| + |\cos x| \le \sqrt{2}$

when $\quad y = 1, x = \pm 2$

So, the point of intersection are (2, 1) and (−2, 1)

Now, $\quad y = 1 \Rightarrow \dfrac{dy}{dx} = 0$

$$\Rightarrow \quad m_1 = \left(\frac{dy}{dx}\right)_P = 0$$

Also, $\quad x^2 + y^2 = 5$

$$\Rightarrow \quad 2x + 2y\frac{dy}{dx} = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = -\frac{x}{y}$$

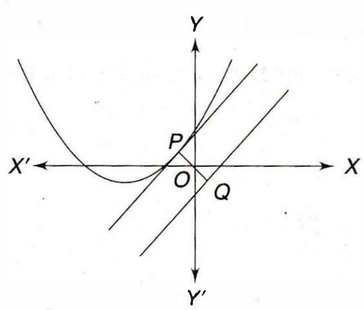
$$\Rightarrow \quad m_2 = \left(\frac{dy}{dx}\right)_P = -2$$

Let θ be the angle between them

Then $\quad \tan\theta = \left|\dfrac{m_2 - m_1}{1 + m_1 \cdot m_2}\right| = \left|\dfrac{-2-0}{1+0}\right| - 2$

$$\Rightarrow \quad \theta = \tan^{-1}(2)$$

43.



The given curves are $y = x^2 + 3x + 2$ and

$y = x - 2$

$$\Rightarrow \quad \frac{dy}{dx} = 2x + 3 \text{ and } \frac{dy}{dx} = 1$$

Since the tangents are parallel, so their slopes are same.

Therefore, $2x + 3 = 1$

$\Rightarrow \quad x = 1$

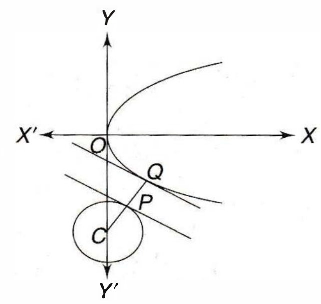
when $x = 1$, then y is 6.

Thus, the point lies on the curve is (1, 6)

Hence, the length of the shortest distance

$$= \left|\frac{1 - 6 + 2}{\sqrt{1^2 + 1^2}}\right| = \frac{3}{\sqrt{2}}$$

44.



The given curves are $y^2 = 4x$ and

$$x^2 + y^2 - 12x + 31 = 0.$$

In this case, tangent at P on the parabola is parallel to the tangent at Q on the circle.

So their slopes are same.

Thus, $\quad \dfrac{4}{2y} = \dfrac{6-x}{y}$

$\Rightarrow \quad x = 4$

when $\quad x = 4$, then $y = -4$

So, the point on the parabola is (4, −4)

Now shortest distance $= PQ$

$= CP - CQ$

$= 2\sqrt{5} - \sqrt{5}$

$= \sqrt{5}$

45. The given curves are $y^2 = x^3$ and

$$9x^2 + 9y^2 - 30y + 16 = 0.$$

In this case, tangent at P on the curve $y^2 = x^3$

So their slopes are same.

Now, $\quad \dfrac{3x^2}{2y} = -\dfrac{3x}{3y-5}$

$$\Rightarrow \quad y = \frac{5x}{3x+2}$$

Put the value of y in $y^2 = x^3$, we get,

$$\frac{25x^2}{(3x+2)^2} = x^3$$

$\Rightarrow \quad 9x^3 + 12x^2 + 4x - 25 = 0$

$\Rightarrow \quad x = 1.$

when $\quad x = 1$, then $y = 1$

So, the point is (1, 1)

Now, the equation of the normal to the curve

$y^2 = x^3$ at (1, 1) is $y - 1 = -\dfrac{2}{3}(x-1)$

$\Rightarrow \quad 2x + 3y = 9$

On solving, we get, the point on the ellipse is

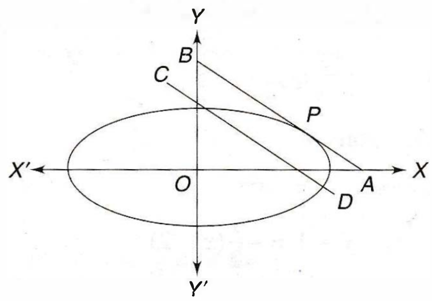
$$\left(\frac{3}{\sqrt{13}}, \frac{5\sqrt{13}-6}{3\sqrt{13}}\right)$$

Hence, the required shortest distance

$$= \sqrt{\left(\frac{3}{\sqrt{13}} - 1\right)^2 + \left(\frac{5\sqrt{13}-6}{3\sqrt{13}}\right)^2}$$

$$= \sqrt{\frac{1}{13}(110 - 30\sqrt{3})}.$$

46.



The given curve is $x^2 + 2y^2 = 6$

$\Rightarrow \quad \frac{x^2}{6} + \frac{y^2}{3} = 1$

since the tangent at P to the ellipse is parallel to the given line, so their slopes are same.

Now, $\quad -\frac{x}{2y} = -1$

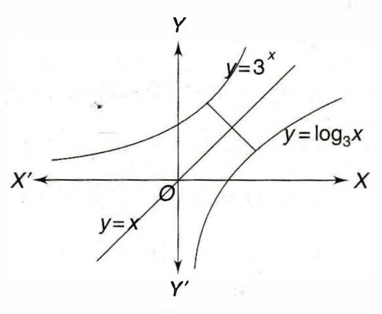
On solving, we get, $6y^2 = 6$

$\Rightarrow \quad y^2 = 1$

$\Rightarrow \quad y = \pm 1$

So, the point can be either (2, 1) or (–2, –1).

47.



The given curves are $y = 3^x$ and $y = \log_3 x$.
Clearly, $y = \log_3 x$ is the image of the curve $y = 3^x$ with respect to the line $y = x$.
Therefore, $3^x \cdot \log 3 = 1$

$\Rightarrow \quad 3^x = \frac{1}{\log 3} = (\log 3)^{-1}$

$\Rightarrow \quad x = \log_3(\log 3)^{-1} = -\log_3(\log 3)$

when $\quad x = -\log_3(\log 3)$, then $y = \frac{1}{\log 3}$

Thus, the point $\left(-\log_3(\log 3), \frac{1}{\log 3}\right)$ lies on the curve $y = 3^x$.

Since the curve $y = \log_3 x$ is the image of the curve $y = 3^x$ with respect to the line $y = x$, so the point on the curve $y = \log_3 x$ is

$$\left(\frac{1}{\log 3}, -\log_3(\log 3)\right)$$

Hence, the shortest distance

$$= \sqrt{\left(\frac{1}{\log 3} + \log_3(\log 3)\right)^2 + \left(\log_3(\log 3) + \frac{1}{\log 3}\right)^2}$$

$$= \sqrt{2}\left(\frac{1}{\log 3} + \log_3(\log 3)\right)$$

$$= \sqrt{2}\left(\frac{1 + \log(\log 3)}{\log 3}\right)$$

$$= \sqrt{\left(\frac{1}{\log 3} + \log_3(\log 3)\right)^2 + \left(\log_3(\log 3) + \frac{1}{\log 3}\right)^2}$$

$$= \sqrt{2}\left(\frac{1}{\log 3} + \log_3(\log 3)\right)$$

$$= \sqrt{2}\left(\frac{1 + \log(\log 3)}{\log 3}\right)$$

48. The given curves are $y = x^2 + x + 1$ and

$$y = x^2 - 5x + 6$$

Let the common tangent be $y = ax + b$.
On solving with both the given curves, we have,
$ax + b = x^2 + x + 1$ and $ax + b = x^2 - 5x + 6$

$\Rightarrow \quad x^2 + (1 - a)x + (1 - b) = 0$

and $\quad x^2 + (5 + a)x + (6 - b) = 0$

Since they have a common tangent, so the given equations have equal roots.

Thus, $\quad D = 0$

$\Rightarrow \quad (1 - a)^2 - 4(1 - b) = 0$

and $\quad (5 + a)^2 - 4(6 - b) = 0$

$\Rightarrow \quad a^2 - 2a + 4b - 3 = 0$

and $\quad a^2 + 10a + 4b + 1 = 0$

$\Rightarrow \quad a = -1/3$ and $b = 5/9$.

Hence, the equation of the common tangent is

$$3x + 9y = 5.$$

49. The given curves are $y = 3x^2$ and $y = 2x^3 + 1$

$$\frac{dy}{dx} = 6x \text{ and } \frac{dy}{dx} = 6x^2$$

Since the given curves have common tangent, so their slopes are same.

Thus, $6x = 6x^2$.

$\Rightarrow \quad x = 0$ and 1.

when $x = 0$, $y = 0$ and when $x = 1$, $y = 3$.

So, the points are (0, 0) and (1, 3).

Hence, the equations of the common tangents are

$$y = 0 \text{ and } y - 1 = 6(x - 1) = 6x - 6$$

$$\Rightarrow \quad y = 0 \text{ and } y = 6x - 5.$$

50. The given curves are $y = 6 - x - x^2$ and $y = 1 + \frac{3}{x}$

$$\Rightarrow \quad \frac{dy}{dx} = -1 - 2x \text{ and } \frac{dy}{dx} = -\frac{3}{x^2}$$

Since the given curves have their common tangent,

so $\quad -\frac{3}{x^2} = -1 - 2x$

$$\Rightarrow \quad 2x^3 + x^2 - 3 = 0$$

$$\Rightarrow \quad x = 1$$

when $\quad x = 1, y = 6 - 1 - 1 = 4$

So, the point is (1, 4)

Hence, the equation of the common tangent is

$$y - 4 = -3(x - 1)$$

$$\Rightarrow \quad y - 4 = -3x + 3$$

$$\Rightarrow \quad 3x + y = 7.$$

51. The given curve is $y^2 = x(2 - x)^2$

$$\Rightarrow \quad 2y\frac{dy}{dx} = (2 - x)^2 - 2x(2 - x)$$

$$\Rightarrow \quad 2y\frac{dy}{dx} = 4 - 4x + x^2 - 4x + 2x^2$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{3x^2 - 8x + 4}{2y}$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{(1,1)} = \frac{3 - 8 + 4}{2} = -\frac{1}{2}$$

Hence, the equation of the tangent is

$$y - 1 = -\frac{1}{2}(x - 1)$$

$$\Rightarrow \quad 2y - 2 = -x + 1$$

$$\Rightarrow \quad x + 2y = 3$$

On solving, the given curve and the tangent, we get,

$$y^2 = (3 - 2y)(2 - 3 + 2y)^2$$

$$\Rightarrow \quad y^2 = (3 - 2y)(2y - 1)^2$$

$$\Rightarrow \quad 8y^3 - 19y^2 + 14y - 3 = 0$$

$$\Rightarrow \quad (y - 1)(8y^2 - 11y + 3) = 0$$

$$\Rightarrow \quad (y - 1)^2(8y - 3) = 0$$

$$\Rightarrow \quad y = 1, 3/8$$

when $y = 3/8$, $x = 3 - 2y = 3 - \frac{3}{4} = \frac{9}{4}$

Hence, the point is $\left(\frac{9}{4}, \frac{3}{8}\right)$.

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53.

54. Given $\quad x = \sec^2\theta$ and $y = \cot\theta$

$$\frac{dx}{d\theta} = 2\sec^2\theta\tan\theta \text{ and } \frac{dy}{d\theta} = -\operatorname{cosec}^2\theta$$

Thus, $\quad \frac{dy}{dx} = -\frac{\operatorname{cosec}^2\theta}{2\sec^2\theta\tan\theta} = -\frac{1}{2}\cot^3\theta$

Now, $\quad \left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{4}} = -\frac{1}{2}$

So, the point P is (2, 1)

Equation of the tangent at P is

$$y - 1 = -\frac{1}{2}(x - 2)$$

$$\Rightarrow \quad 2y - 2 = -(x - 2) = -x + 2$$

$$\Rightarrow \quad x + 2y = 4 \quad \text{...(i)}$$

Eliminating 'θ' between $x = \sec^2\theta$ & $y = \cot\theta$

we get, $x - \frac{1}{y^2} = 1 \quad$ ...(ii)

On solving (i) and (ii), we get,

$$\Rightarrow \quad 4 - 2y - \frac{1}{y^2} = 1$$

$$\Rightarrow \quad 4y^2 - 2y^3 - 1 = y^2$$

$$\Rightarrow \quad 3y^2 - 2y^3 - 1 = 0$$

$$\Rightarrow \quad 2y^3 - 3y^2 + 1 = 0$$

$$\Rightarrow \quad 2y^3 - 2y^2 - y^2 + y - y + 1 = 0$$

$$\Rightarrow \quad 2y^2(y - 1) - y(y - 1) - (y - 1) = 0$$

$$\Rightarrow \quad (y - 1)(2y^2 - y - 1) = 0$$

$$\Rightarrow \quad (y - 1) = 0, (2y^2 - y - 1) = 0$$

$$\Rightarrow \quad (y - 1) = 0, (2y^2 - 2y + y - 1) = 0$$

$$\Rightarrow \quad (y - 1) = 0, (2y(y - 1) + (y - 1)) = 0$$

$$\Rightarrow \quad (y - 1) = 0, (y - 1)(2y + 1) = 0$$

$$\Rightarrow \quad y = 1, -\frac{1}{2}$$

$$\Rightarrow \quad y = -\frac{1}{2}$$

when $y = -\frac{1}{2}$, then $x = 4 - 2y = 4 + 2\cdot\frac{1}{2} = 5$

Thus, the point Q is $\left(5, -\frac{1}{2}\right)$

Now, the length of PQ

$$= \sqrt{(2-5)^2 + \left(1 + \frac{1}{2}\right)^2}$$

$$= \sqrt{9 + \frac{9}{4}}$$

$$= \sqrt{\frac{45}{4}}$$

$$= \frac{3\sqrt{5}}{2}$$

55. The given curve is $y^2 = 4ax$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{4a}{2y} = \frac{2a}{y}$$

Now, $\left(\frac{dy}{dx}\right)_P = \frac{2a}{2at} = \frac{1}{t}$

(i) The length of the tangent

$$= y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$= 2at\sqrt{1 + t^2}$$

(ii) The length of the sub-tangent

$$= y \cdot \frac{dx}{dy}$$

$$= 2at \times t = 2at^2$$

(iii) The length of the normal

$$= y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$= 2at \times \sqrt{1 + \frac{1}{t^2}}$$

$$= 2a\sqrt{t^2 + 1}$$

(iv) The length of the sub-normal

$$= y \cdot \frac{dy}{dx}$$

$$= 2at \times \frac{1}{t}$$

$$= 2a.$$

56. The given curve is $y = be^{x/a}$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{b}{a}e^{x/a}$$

Let the point be (x, y)

Now, the length of the sub-tangent

$$= y \cdot \frac{dx}{dy}$$

$$= be^{x/a} \times \frac{a}{be^{x/a}}$$

$$= a$$

$$= \text{constant.}$$

57. The given curves are $x = a(\theta - \sin\theta)$,

$$y = a(1 - \cos\theta)$$

$$\Rightarrow \quad \frac{dx}{d\theta} = a(1 - \cos\theta) \text{ and } \frac{dy}{d\theta} = a\sin\theta$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{a\sin\theta}{a(1 - \cos\theta)}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2\sin^2\left(\frac{\theta}{2}\right)}$$

$$\Rightarrow \quad \frac{dy}{dx} = \cot\left(\frac{\theta}{2}\right)$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{\theta = \frac{\pi}{2}} = \cos\left(\frac{\pi}{4}\right) = 1$$

when $\theta = \frac{\pi}{4}$, $y = a\left(\frac{1-1}{\sqrt{2}}\right)$

The length of the tangent

$$= y\sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

$$= a\left(1 - \frac{1}{\sqrt{2}}\right) \times \sqrt{2}$$

$$= a(\sqrt{2} - 1)$$

The length of the normal

$$= y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$= a\left(1 - \frac{1}{\sqrt{2}}\right) \times \sqrt{2}$$

$$= a(\sqrt{2} - 1)$$

58. The given curve is $y^2 = x^3$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{3x^2}{2y}$$

$$\Rightarrow \quad \left(\frac{dy}{dx}\right)_{(4,8)} = \frac{3.16}{2.8} = 3$$

Hence, the length of the sub-normal

$$= y \cdot \frac{dy}{dx} = 8.3 = 24$$

59. The given curve is $y = \frac{c}{2}\left(e^{\frac{x}{c}} + e^{-\frac{x}{c}}\right)$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{c}{2}\left(\frac{1}{c}e^{\frac{x}{c}} - \frac{1}{c}e^{-\frac{x}{c}}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{2}\left(e^{\frac{x}{c}} - e^{-\frac{x}{c}}\right)$$